AS1109 - Student Notes

Quadric Surfaces and Solids of Revolution

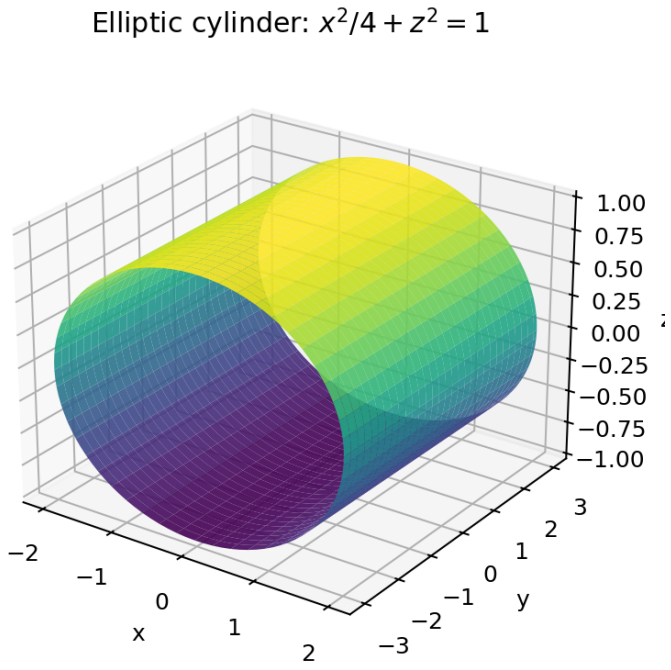
1. Quadric Surfaces

Cylinder rule

If one coordinate is missing from a three-dimensional equation, the same plane curve is repeated parallel to that missing axis.
Example:

$$\frac{x^2}{4} + z^2 = 1$$

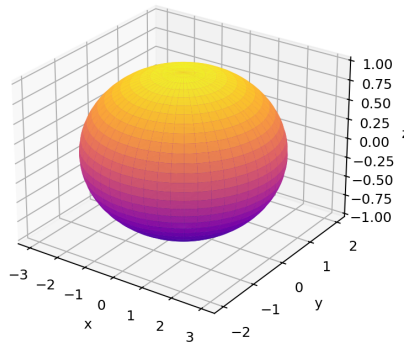
contains no y, so it is an elliptic cylinder parallel to the y-axis.



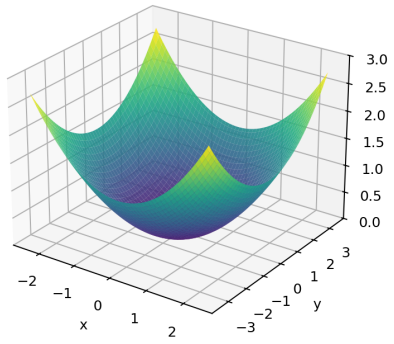
Recognition table

Surface	Representative form	Recognition clue
Ellipsoid	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$	all positive squares; bounded
Elliptic paraboloid	$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$	one unsquared variable; bowl
Hyperbolic paraboloid	$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$	opposite signs; saddle
Elliptic cone	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$	homogeneous quadratic; double cone
One-sheet hyperboloid	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$	one negative square; connected
Two-sheet hyperboloid	$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$	one positive square; two pieces

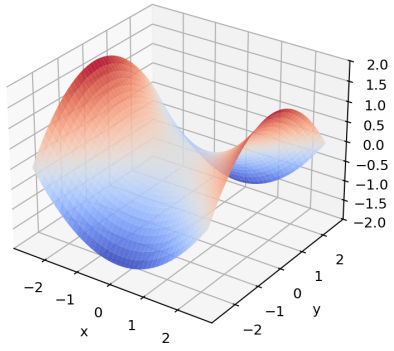
Ellipsoid: $x^2/9 + y^2/4 + z^2 = 1$



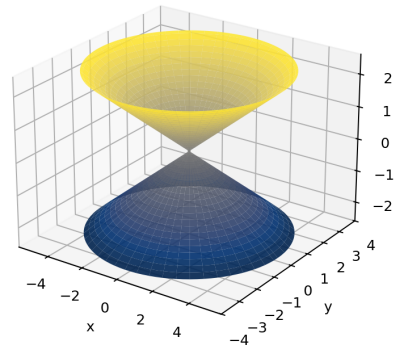
Elliptic paraboloid: $z = x^2/4 + y^2/9$



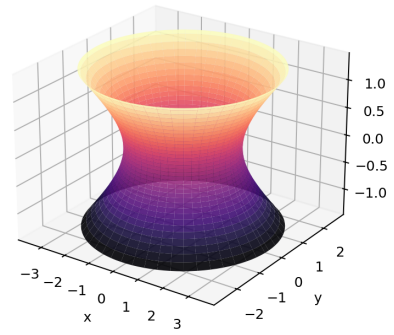
Hyperbolic paraboloid: $z = x^2/4 - y^2/4$



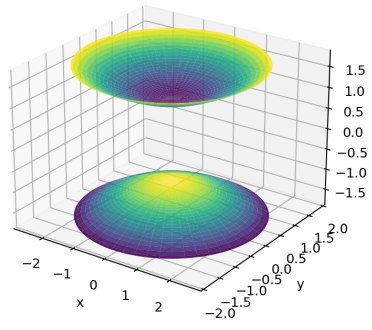
Elliptic cone: $x^2/4 + y^2/1.96 = z^2$



One-sheet hyperboloid: $x^2/4 + y^2/2.25 - z^2 = 1$



Two-sheet hyperboloid: $-x^2/2.89 - y^2/1.44 + z^2 = 1$



Trace method

Set one coordinate equal to a constant:

$$x = c, \qquad y = c, \qquad z = c.$$

The resulting 2D curves are called traces. They are often more useful than memorising a 3D picture.

Solved example 1: elliptic paraboloid

Identify

$$z = \frac{x^2}{4} + \frac{y^2}{9}$$

using traces.

Detailed solution

For $y = 0$,

$$z = \frac{x^2}{4},$$

a parabola.
For $x = 0$,

$$z = \frac{y^2}{9},$$

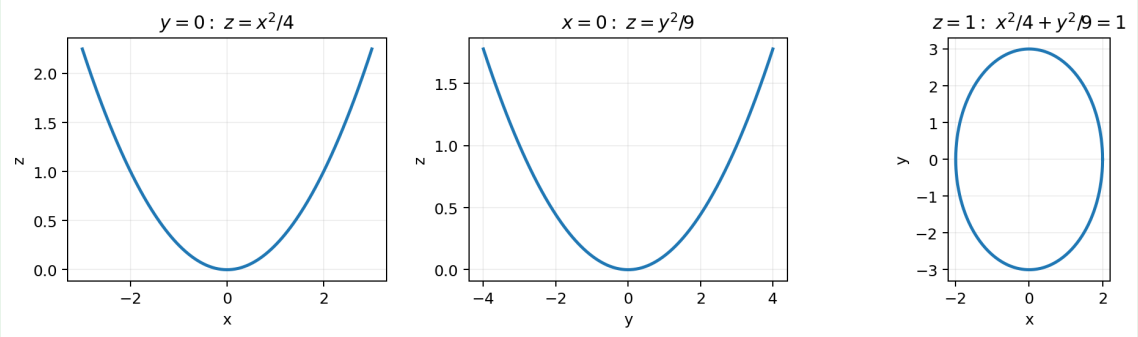
another upward-opening parabola.
For $z = c > 0$,

$$\frac{x^2}{4} + \frac{y^2}{9} = c$$

or

$$\frac{x^2}{4c} + \frac{y^2}{9c} = 1,$$

an ellipse.
Therefore the surface is an elliptic paraboloid opening in the positive z-direction.



Solved example 2: hyperbolic paraboloid

Identify

$$z = \frac{x^2}{4} - \frac{y^2}{4}.$$

Detailed solution

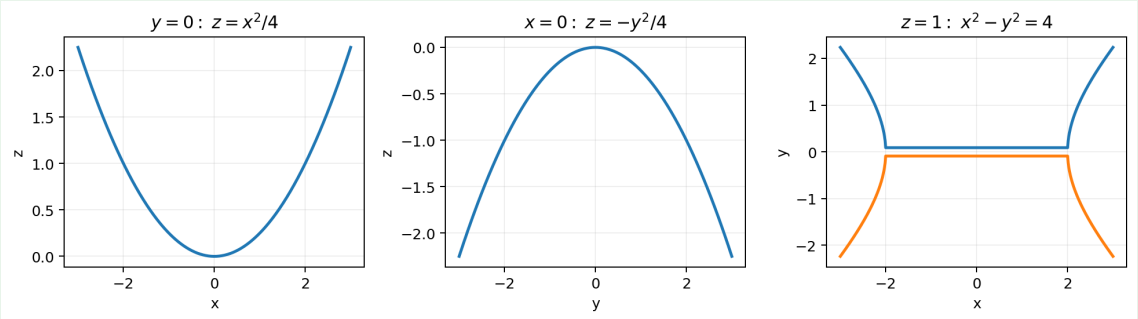
For $y = 0$,

$$z = \frac{x^2}{4},$$

an upward parabola.
For $x = 0$,

$$z = -\frac{y^2}{4},$$

a downward parabola.
These opposite vertical traces create a saddle. Hence the surface is a hyperbolic paraboloid.



Engineering shapes

Elliptic paraboloids model idealised reflector dishes; hyperbolic paraboloids occur in saddle-shaped shell structures; hyperboloid-like forms appear in structural and tower geometries.

Common mistake

A reliable identification gives

$$\boxed{\text{standard form/sign pattern + at least one or two confirming traces}}.$$

Practice

1. Identify $x^2 + y^2 = 16$ in three-dimensional space.
2. Identify $\frac{x^2}{9} + \frac{y^2}{4} - z^2 = 1$.
3. For $z = x^2 + 4y^2$, describe the traces $x = 0$, $y = 0$, and $z = 1$.
4. How does the sign pattern distinguish a one-sheet from a two-sheet hyperboloid?

2. Surface Area of Solids of Revolution

Main idea

A short curve element of arc length ds, at distance r from the axis of rotation, sweeps approximately a thin frustum:

$$\boxed{dS \approx 2\pi r\, ds}.$$

For $y = f(x)$,

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$$

Hence, about the x-axis,

$$\boxed{S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx}.$$

For $x = g(y)$, about the y-axis,

$$\boxed{S = \int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy}.$$

Common mistake

The factor r always means **distance to the axis of rotation**. It is not automatically x or y.

Solved example 3: hemisphere surface area

Rotate

$$y = \sqrt{4 - x^2}, \qquad 0 \le x \le 2,$$

about the x-axis.

Detailed solution

$$y' = -\frac{x}{\sqrt{4 - x^2}}.$$

Then

$$1 + (y')^2 = 1 + \frac{x^2}{4 - x^2} = \frac{4}{4 - x^2}.$$

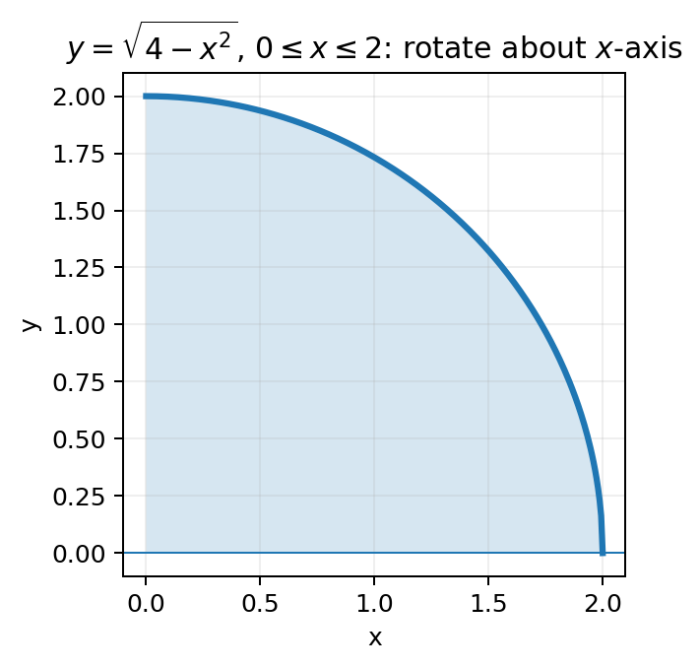
Thus

$$\sqrt{1 + (y')^2} = \frac{2}{\sqrt{4 - x^2}} = \frac{2}{y}.$$

Therefore

$$\begin{aligned} S &= \int_0^2 2\pi y \sqrt{1 + (y')^2}\, dx \\ &= \int_0^2 4\pi\, dx \\ &= \boxed{8\pi}. \end{aligned}$$

This is half the area $4\pi(2)^2 = 16\pi$ of a sphere of radius 2.



Solved example 4: rotation about the y-axis

Rotate

$$y = x^2, \qquad 0 \le x \le 1,$$

about the y-axis.

Detailed solution

The radius is

$$r = x.$$

Also

$$\frac{dy}{dx} = 2x.$$

Hence

$$S = 2\pi \int_0^1 x\sqrt{1 + 4x^2}\, dx.$$

Let

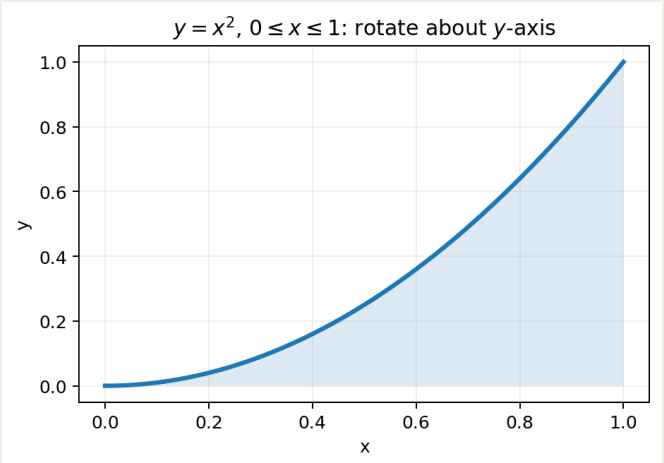
$$u = 1 + 4x^2, \qquad du = 8x\, dx.$$

Then

$$S = \frac{\pi}{4} \int_1^5 u^{1/2}\, du = \frac{\pi}{6} \left[u^{3/2}\right]_1^5.$$

Therefore

$$\boxed{S = \frac{\pi}{6}(5\sqrt{5} - 1)}.$$



Surface-area setup checklist

$$\boxed{\text{curve} \to \text{axis} \to \text{radius} \to ds \to S = \int 2\pi r \, ds}$$

Practice

1. Set up the surface area generated by rotating $y = x^3$, $0 \le x \le 1$, about the x-axis.
2. Set up the same curve about the y-axis.
3. Why does surface area use ds rather than simply dx?

3. Volumes of Solids of Revolution

Slicing principle

A thin slice of thickness dx has approximate volume

$$dV \approx A(x)\,dx.$$

Therefore

$$\boxed{V = \int A(x)\,dx}.$$

For a disk:

$$\boxed{A = \pi R^2}.$$

For a washer:

$$\boxed{A = \pi(R^2 - r^2)}.$$

Common mistake

For a washer,

$$\boxed{\pi(R^2 - r^2)}$$

is correct. Do not write $\pi(R - r)^2$.

Solved example 5: disk method

Rotate

$$y = \sqrt{4 - x}, \qquad 0 \le x \le 4,$$

about the x-axis.

Detailed solution

The radius is

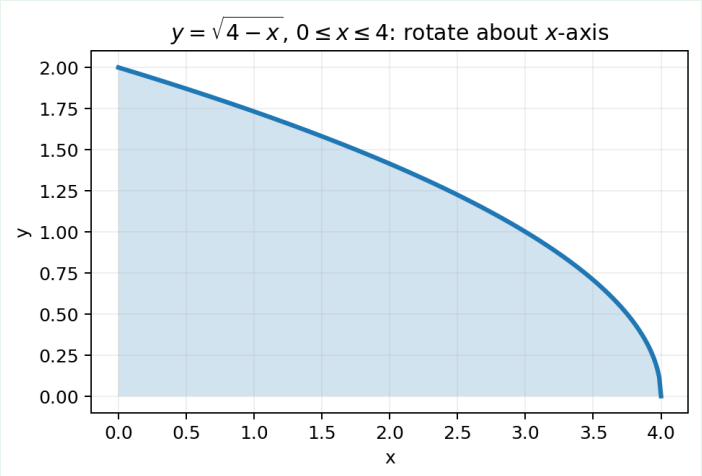
$$R(x) = \sqrt{4 - x}.$$

Therefore

$$A(x) = \pi R^2 = \pi(4 - x).$$

Thus

$$\begin{aligned} V &= \pi \int_0^4 (4 - x)\,dx \\ &= \pi \left[4x - \frac{x^2}{2}\right]_0^4 \\ &= \boxed{8\pi}. \end{aligned}$$



Solved example 6: washer method between two curves

The region between

$$y = \sqrt{x} \qquad \text{and} \qquad y = x,$$

for $0 \le x \le 1$, is rotated about the x-axis.

Detailed solution

For $0 < x < 1$,

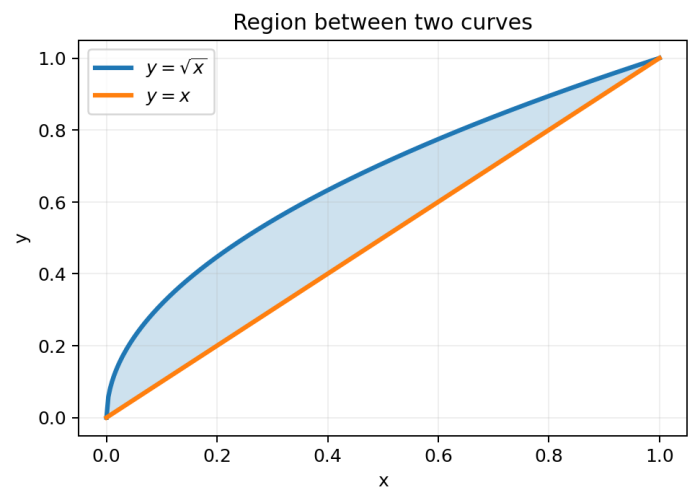
$$\sqrt{x} > x.$$

Hence

$$R(x) = \sqrt{x}, \qquad r(x) = x.$$

Thus

$$\begin{aligned} V &= \pi \int_0^1 \left(R^2 - r^2\right)\, dx \\ &= \pi \int_0^1 (x - x^2)\, dx \\ &= \pi \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 \\ &= \boxed{\frac{\pi}{6}}. \end{aligned}$$



Solved example 7: rotation about the y-axis

The region bounded by

$$x = y^2, \qquad x = 4, \qquad 0 \le y \le 2$$

is rotated about the y-axis.

Detailed solution

Use horizontal slices, so integrate with respect to y.
Outer radius:

$$R = 4.$$

Inner radius:

$$r = y^2.$$

Therefore

$$\begin{aligned} V &= \pi \int_0^2 (16 - y^4)\, dy \\ &= \pi \left[16y - \frac{y^5}{5}\right]_0^2 \\ &= \boxed{\frac{128\pi}{5}}. \end{aligned}$$

Geometry-first routine

$$\boxed{\text{draw region} \to \text{mark axis} \to \text{draw perpendicular slice} \to \text{label } R, r \to \text{integrate}}$$

Engineering interpretation

Disk/washer integrals model volumes of tanks, turned components, hollow shafts, nozzles and cavities. If density is constant,

$$\boxed{m = \rho V}.$$

Practice

1. Find the volume generated by rotating $y = 2x$, $0 \le x \le 1$, about the x-axis.
2. Set up the volume when the region between $y = 2$ and $y = x^2$, $-\sqrt{2} \le x \le \sqrt{2}$, is rotated about the x-axis.
3. For a rotation about the y-axis, explain when integrating with respect to y is the natural choice.